

For example,

$$\frac{5}{8} \times \frac{2}{3} = \frac{5 \times 2}{8 \times 3} = \frac{10}{24} \quad \left(= \frac{5}{12} \text{ in simplest form} \right)$$

$$12 \times \frac{5}{8} = \frac{12}{1} \times \frac{5}{8} = \frac{12 \times 5}{1 \times 8} = \frac{60}{8} \quad \left(= \frac{15}{2} = 7\frac{1}{2} \right)$$

To divide by a fraction, multiply by the **reciprocal** of the fraction. For example,

$$35 \div \frac{5}{8} = \frac{35}{1} \times \frac{8}{5} = \frac{35 \times 8}{1 \times 5} = \frac{7 \times 8}{1} = 56$$

$$\frac{5}{8} \div 3\frac{3}{4} = \frac{5}{8} \div \frac{15}{4} = \frac{5}{8} \times \frac{4}{15} = \frac{5 \times 4}{8 \times 15} = \frac{20}{120} \quad \left(= \frac{1}{6} \right)$$

f $x\%$ is short for $\frac{x}{100}$. 64% means $\frac{64}{100}$.

To change a fraction to an equivalent percentage, multiply the fraction by 100. For example, $\frac{5}{8}$ as a percentage $= \frac{5}{8} \times 100\% = \frac{125}{2}\% = 62\frac{1}{2}\%$.

g To change a fraction to a **decimal fraction**, divide the numerator by the denominator.

For example,

$$\begin{array}{r} \frac{5}{8} = 0.625 \\ 8 \overline{) 5.000} \\ \underline{48} \\ 20 \\ \underline{16} \\ 40 \\ \underline{40} \\ 0 \end{array}$$

When adding or subtracting decimals, write the numbers in a column with the decimal points exactly under each other. For example: Add 2.29, 0.084 and 4.3, then subtract the result from 11.06.

$$\begin{array}{r} 2.29 \\ 0.084 \\ + 4.3 \\ \hline 6.674 \end{array} \quad \begin{array}{r} 11.06 \\ - 6.674 \\ \hline 4.386 \end{array}$$

To multiply decimals, first ignore the decimal points and multiply the given numbers as if they are whole numbers. Then place the decimal point so that the answer has as many digits after the point as there are in the given numbers together. For example, 0.08×0.3 :

$$8 \times 3 = 24$$

There are three digits after the decimal points in the given numbers, so $0.08 \times 0.3 = 0.024$.

To divide by decimals, make an equivalent division such that the divisor is a whole number. For example, $5.6 \div 0.07$:

$$5.6 \div 0.07 = \frac{5.6}{0.07} = \frac{5.6 \times 100}{0.07 \times 100} = \frac{560}{7} = 80$$

h Numbers may be positive or negative. Positive and negative numbers are called **directed numbers**. Directed numbers can be shown on a **number line** (Fig. P4).

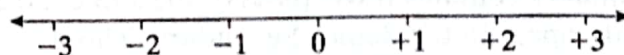


Fig. P4

The following examples show how to add, subtract, multiply and divide directed numbers.

addition

$$(+8) + (+3) = +11$$

$$(+8) + (-3) = +5$$

$$(-8) + (+3) = -5$$

$$(-8) + (-3) = -11$$

subtraction

$$(+9) - (+4) = +5$$

$$(+9) - (-4) = +13$$

$$(-9) - (+4) = -13$$

$$(-9) - (-4) = -5$$

multiplication

$$(+2) \times (+7) = +14$$

$$(+2) \times (-7) = -14$$

$$(-2) \times (+7) = -14$$

$$(-2) \times (-7) = +14$$

division

$$(+6) \div (+3) = +2$$

$$(+6) \div (-3) = -2$$

$$(-6) \div (+3) = -2$$

$$(-6) \div (-3) = +2$$

An **integer** is any positive or negative whole number as shown in Fig. P4 above.

i The number 3.7×10^4 is in **standard form**. The first part of the product is between 1 and 10, and the second is a power of 10.

j When **rounding off** numbers, the digits 1, 2, 3, 4 are rounded down, and the digits 5, 6, 7, 8, 9 are rounded up. For example,

$$\begin{aligned} 3425 &= 3430 \text{ to 3 significant figures} \\ &= 3400 \text{ to the nearest hundred} \end{aligned}$$

$$\begin{aligned} 7.283 &= 7.28 \text{ to 2 decimal places} \\ &= 7.3 \text{ to 1 decimal place} \\ &= 7 \text{ to the nearest whole number} \end{aligned}$$

When calculating from 4-figure tables, it is usual to round numbers to 3 s.f.

k **Binary numbers** use a **base two** place value system. For example,

$$\begin{aligned} 110101_{\text{two}} &= 1 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + \\ &\quad 1 \times 2^2 + 0 \times 2^1 + 1 \\ &= 32 + 16 + 0 + 4 + 0 + 1 = 53_{\text{ten}} \end{aligned}$$

To change a base ten number to base two, divide repeatedly by 2 until further division is impossible. The remainders in each line of working, reading upwards, give the required number. For example, Change 29_{ten} to a binary number.

		rem.
2	29	
2	14	1
2	7	0
2	3	1
2	1	1
	0	1

$29_{ten} = 11\ 101_{two}$

- 1 A **rational number** is any number in the form $\frac{a}{b}$, where a and b are whole numbers. For example, $\frac{1}{9}$, $\frac{15}{1}$, $\frac{33}{8}$, $\frac{5}{5}$ are rational numbers.

Non-rational (or **irrational**) **numbers** cannot be expressed as exact fractions.

For example, $\sqrt{2}$ ($=1.414\ 213\dots$) and π ($=3.141\ 592\dots$) are non-rational numbers.

Detailed coverage of number and numeration is given in the JSS course as follows: Book 1, Chapters 1, 2, 4, 9, 12, 23; Book 2, Chapters 1, 2, 4, 9, 10, 15, 18; and Book 3, Chapters 1, 5, 11, 20, 21.

Review test 1 (Number and numeration)

Allow 35 minutes for this test. Use the answers at the back of the book to check your work. If you do not understand why some of your answers were incorrect, ask a friend or your teacher. Then try Review test 2.

- Express each of the following as a product of factors. Hence find their square roots.
a 14 400 b 324 c 2 025 d 7 056
- Express the following in standard form.
a 7 000 000 b 80 000
c 0.0009 d 0.000 002
- Round off the following to
i 1 s.f., ii 2 s.f., iii 3 s.f.
a 7 607 b 42 824
c 31.499 d 0.007 407

- Round off the following to
i 1 d.p., ii 2 d.p., iii 3 d.p.
a 23.7016 b 0.0094
c 8.8535 d 0.0077
- Round off the following to the nearest
i whole number, ii tenth, iii hundredth.
a 5.049 b 59.509
- Simplify the following.
a $(-9) \times (-2)$ b $33 \times (-3)$
c $-51 \div 17$ d $(-35) \div (-5)$
- A school has 988 students. The average mass of a student is approximately 51 kg. Estimate, to 1 significant figure, the total mass in tonnes of all the students in the school.
- Divide the following quantities in the given ratios.
a ₦2 800 in the ratio 1:3
b 800 mm in the ratio 3:2
c 88 bananas in the ratio 6:5
- Find the discount price if
a 10% is given on a cost price of ₦600
b 25% is given on a cost price of ₦6 600
- Use tables to find
a 170^2 b $2\ 900^2$
c $\sqrt{5\ 446}$ d $\sqrt{804}$
- a Convert to base ten:
i $10\ 100_{two}$, ii $110\ 011_{two}$
b Convert to base two: i 23_{ten} , ii 60_{ten}
- Which of the following numbers are non-rational?
 3 , 3.3 , 3.3 , $3\frac{1}{3}$, $\sqrt{3}$, 3.142 , π

Review test 2 (Number and numeration)

Allow 35 minutes for this test. Use the answers at the back of the book to check your work. If you do not understand why some of your answers were incorrect, ask a friend or your teacher.

- Express each of the following as a product of factors. Hence find their square roots.
a 4 900 b 11 664
c 1 600 d 1 936

2 Express the following in standard form.

- a 5000 b 600 000 000
c 0.00008 d 0.005

3 Round off the following to

- i 1 s.f., ii 2 s.f., iii 3 s.f.

- a 657.63 b 9.0175
c 69.487 d 0.08855

4 Round off the following to

- i 1 d.p., ii 2 d.p., iii 3 d.p.

- a 46.3399 b 5.0275
c 0.0493 d 0.0084

5 Round off the following to the nearest

- i whole number, ii tenth, iii hundredth.

- a 0.625 b 8.385

6 Simplify the following.

- a $(-61) \times (-8)$
b -8×10
c $68 \div -4$
d $(-72) \div (-8)$

7 A typist can type about 29 words per minute. Estimate, to 1 significant figure, how long it will take her to copy-type a text that is 1006 words long.

8 Divide the following quantities in the given ratios.

- a ₦900 in the ratio 3:2
b 90 ml in the ratio 4:1
c 84 eggs in the ratio 1:5

9 Find the discount price if

- a 5% is given on a cost price of ₦4000
b 20% is given on a cost price of ₦680

10 Use tables to find:

- a 57^2 b 1050^2
c $\sqrt{8005}$ d $\sqrt{664}$

11 a Convert to base ten:

- i 111001_{two} ii 101110_{two}

b Convert to base two: i 33_{ten} ii 59_{ten}

12 Which of the following numbers are rational?

0.9, 0.9, 9, $\sqrt{9}$, $\sqrt{90}$, $\sqrt{900}$, 9.9.

Algebraic processes

a $3y^2 + 2x - 7x$ is an example of an **algebraic expression**. The letters y and x stand for numbers. $3y^2$, $2x$ and $7x$ are the **terms** of the expression. $3y^2$ is short for $3 \times y \times y$. 3 is the **coefficient** of y^2 . Algebraic terms may be **simplified** by combining **like terms**. Thus, $3y^2 + 2x - 7x = 3y^2 - 5x$ since $2x$ and $7x$ are **like terms** (i.e. both terms in x).

b $3(5x - 2) = 11x$ is an **algebraic sentence** containing an equals sign; it is an **equation** in x . x is the **unknown** of the equation. To **solve an equation** means to find the value of the unknown which makes the equation true. Use the **balance method** to solve equations. For example,

$$3(5x - 2) = 11x$$

Clear brackets.

$$15x - 6 = 11x$$

Subtract $11x$ from both sides.

$$15x - 11x - 6 = 11x - 11x$$

$$4x - 6 = 0$$

Add 6 to both sides.

$$4x - 6 + 6 = 0 + 6$$

$$4x = 6$$

Divide both sides by 4.

$$\frac{4x}{4} = \frac{6}{4}$$

$$x = 1\frac{1}{2}$$

In general, when solving equations, i first clear brackets and fractions, ii using equal additions and/or subtractions, collect unknown terms on one side of the equals sign and known terms on the other, iii where necessary, divide or multiply both sides of the equation by the same number to find the unknown.

c An **inequality** is an algebraic sentence that contains an inequality sign:

$<$ is less than

$\leq$ is less than or equal to

$>$ is greater than

$\geq$ is greater than or equal to

$\neq$ is not equal to

Inequalities are solved in much the same way as equations. However, when both sides of an inequality are multiplied or divided by a negative number, the inequality sign is *reversed*. For example,

$$\text{If } -3a \leq 12$$

divide both sides by -3 and reverse the inequality

$$\text{Then } a \geq -4$$

- d $2x - 5y = 16$ is a **linear equation** with two **variables**, x and y . There are many pairs of values of x and y which **satisfy** this equation (i.e. make the equation true). For example, if $x = 13$ and $y = 2$ or if $x = 8$ and $y = 0$, the equation will be true. Given two equations, such as $2x - 5y = 16$ and $x + 4y = -5$, it is usually possible to find values of x and y which satisfy *both* equations simultaneously (i.e. at the same time). To solve a pair of **simultaneous linear equations**, either use a graphical method or the method of elimination and substitution. For example,

$$2x - 5y = 16 \quad (1)$$

$$x + 4y = -5 \quad (2)$$

Multiply (2) by 2.

$$2x + 8y = -10 \quad (3)$$

Subtract (3) from (1) to eliminate terms in x .

$$-13y = 26$$

$$y = -2$$

Substitute -2 for y in (2) to find x .

$$x + 4(-2) = -5.$$

$$x - 8 = -5$$

$$x = 3$$

The solution is $x = 3$ and $y = -2$.

- e A **graph** of an algebraic sentence is a picture representing the meaning of the sentence. Graphs of equations and inequalities in one variable can be shown on the number line (Fig. P5).

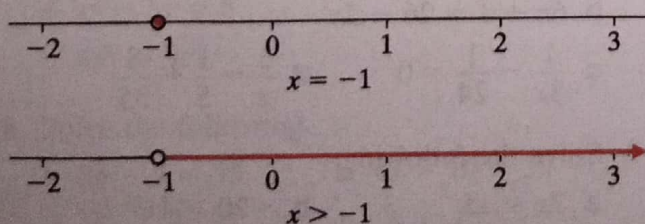


Fig. P5

For graphs connecting two variables, two **axes** are drawn at right angles to each other to give a **cartesian plane**. The horizontal **x -axis** and the vertical **y -axis** cross at their zero-point, the **origin** of the plane. Fig. P6 is the graph of the equation $y = 2x - 3$.

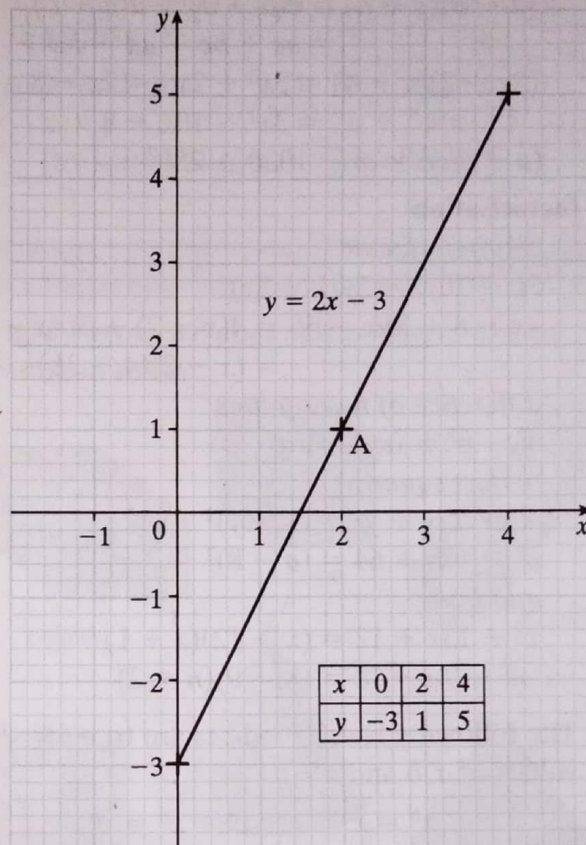


Fig. P6

To draw a straight-line graph, plot at least three points which satisfy the given equation. See the table of values in Fig. P6. At point A in Fig. P6, $x = 2$ and $y = 1$. The **coordinates** of A are A(2, 1). The **order** of the coordinates is important: the **x -coordinate** is given first, the **y -coordinate** second.

Straight-line graphs can be drawn to represent any two connected variables: for example, cost and quantity, distance and time, temperature and time. Straight-line graphs can also be drawn to show conversions between currencies or between marks and percentages.

f Algebraic expressions may be **factorised** or **expanded** in accordance with the basic rules of arithmetic. Some examples are given below.

expansion

$$3(a - 2b) = 3a - 6b$$

$$(5 + 8x)x = 5x + 8x^2$$

$$(a + b)(c + d) = c(a + b) + d(a + b) \\ = ac + bc + ad + bd$$

$$(3x + 2)(x - 4) = 3x^2 + 2x - 12x - 8 \\ = 3x^2 - 10x - 8$$

$$(a - 5b)^2 = a^2 - 10ab + 25b^2$$

factorisation

common factor

$$5y - 10y^2 = 5y(1 - 2y)$$

$$4x - 8 + 3bx - 6b = 4(x - 2) + 3b(x - 2) \\ = (x - 2)(4 + 3b)$$

difference of two squares

$$a^2 - b^2 = (a + b)(a - b)$$

perfect square

$$x^2 + 2xy + y^2 = (x + y)^2$$

$$a^2 - 16a + 64 = (a - 8)^2$$

quadratic

$$x^2 + 13x + 12 = (x + 12)(x + 1)$$

$$n^2 - 7n - 18 = (n - 9)(n + 2)$$

g The following laws of indices are true for all values of a , b and $x \neq 0$.

$$x^a \times x^b = x^{a+b} \quad x^a \div x^b = x^{a-b}$$

$$x^0 = 1 \quad x^{-a} = \frac{1}{x^a}$$

h If x is **directly proportional** to y , then $x \propto y$ or $x = ky$ (where k is a constant). This is an example of **direct variation**.

If x is **inversely proportional** to y , then $x \propto \frac{1}{y}$ or $x = \frac{K}{y}$ (where K is a constant). This is an example of **inverse variation**.

If x is directly proportional to y and inversely proportional to z , then $x \propto \frac{y}{z}$ or

$x = \frac{Cy}{z}$ (where C is a constant). This is an example of **joint variation**.

Detailed coverage of algebraic processes is given in the JSS course as follows: Book 1, Chapters 5, 7, 10, 15, 19; Book 2, Chapters 5, 7, 11, 12, 13, 14, 22; and Book 3, Chapters 2, 3, 6, 9, 10, 13, 14, 19, 23, 24.

Review test 3 (Algebraic processes)

Allow 35 minutes for this test. Use the answers at the back of the book to check your work. If you do not understand why some of your answers were incorrect, ask a friend or your teacher. Then try Review test 4.

1 State the additive inverse of

a -2 **b** $+11$

c -772 **d** $+\frac{5}{9}$

2 State the multiplicative inverse of

a $+5$ **b** $-\frac{3}{8}$

c $+24$ **d** -0.75

3 Remove brackets and simplify.

a $3(3a - b) - 5(a - 4b)$

b $y(2y - 3) - 3(2y - 3)$

4 Expand the following.

a $(6 - x)(3 + x)$

b $(5x - 2y)(2x - 3y)$

5 State the HCF of the following.

a $12x^2$ and $30x^2$ **b** $24xy^2$ and $6x^2y$

6 State the LCM of the following.

a pq and qr **b** $2a^2$ and $7ab$

7 Factorise the following.

a $2a^2 - 9ab$

b $-18pq - 12p$

c $9x + 6 - 3x^2 - 2x$

d $3a - cb - 3b + ac$

e $x^2 - 8x + 16$

f $100 - 49x^2$

8 Solve the following.

a $51 = 3 + 8x$

b $6x + 1 = 26 - 2x$

c $\frac{1}{3x} - \frac{1}{24} = 0$ **d** $\frac{3}{x} = \frac{1}{5} + \frac{8}{35}$

9 Solve the following.

a $2x < 15$

b $-20 > 4x$

c $3x - 8 \leq 5x$

d $8 - 7x > 43 - 2x$

- 10 Solve the following pairs of simultaneous equations.

a $y = 2x - 4$	b $5x - 2y = 14$
$3x + y = 11$	$2x + 2y = 14$
[by substitution]	[by elimination]

Review test 4 (Algebraic processes)

Allow 35 minutes for this test. Use the answers at the back of the book to check your work. If you do not understand why some of your answers were incorrect, ask a friend or your teacher.

- 1 State the additive inverse of

a +8	b -23
c +664	d -0.37

- 2 State the multiplicative inverse of

a 8	b $\frac{1}{10}$
c -12	d -0.8

- 3 Remove brackets and simplify.

a $7x(x - 1) - 5(x - 1)$
b $a(2a + 5b) - b(a + b)$

- 4 Expand the following.

a $(x + 8)(x - 2)$	b $(4x - 7)(3x + 2)$
--------------------	----------------------

- 5 State the HCF of the following.

a $24xy$ and $8xy$	b $18xy^2$ and $12x^2y$
--------------------	-------------------------

- 6 State the LCM of the following.

a $3a$ and $8b$	b $15xy$ and $2x^2y^2$
-----------------	------------------------

- 7 Factorise the following.

a $5a + 5z$
b $2a^2 + 10a$
c $8m - 2 + 4nm - n$
d $6a + bm + 6b + am$
e $x^2 - 5x + 6$
f $4d^2 - 9e^2$

- 8 Solve the following.

a $8g - 7 = 33$	b $4x - 20 = 5x - 22$
c $\frac{3}{10} = \frac{9}{2x}$	d $\frac{5}{2x} - \frac{1}{x} = \frac{1}{6}$

- 9 Solve the following.

a $4x > 20$
b $-11 < 3x$
c $x + 4 \geq 10x - 23$
d $2 - x < 14 - 3x$

- 10 Solve the following pairs of simultaneous equations.

a $y = x + 1$	b $5x + 3y = 1$
$x + y = 3$	$2x + 3y = -5$
[by substitution]	[by elimination]

Geometry and mensuration

- a Fig. P7 gives sketches and names of some common solids.

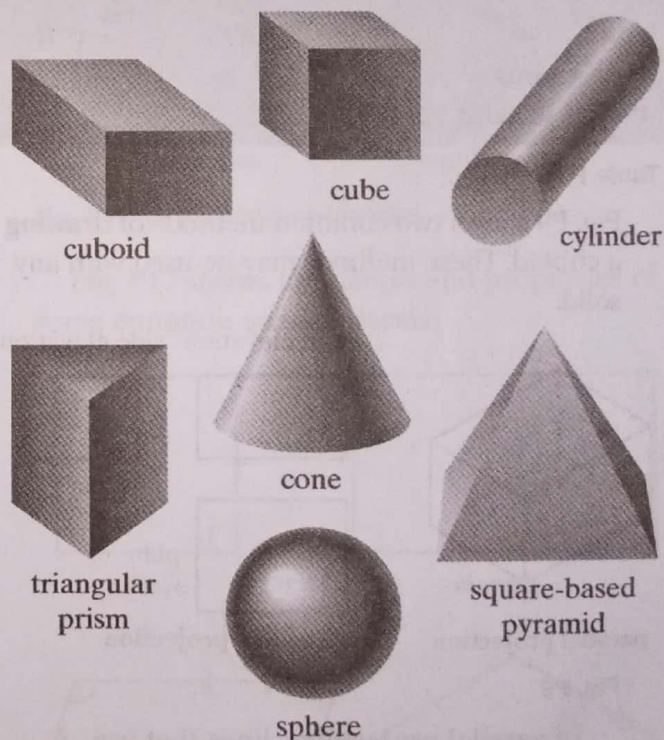


Fig. P7

All solids have **faces**; most solids have **edges** and **vertices** (Fig. P8).

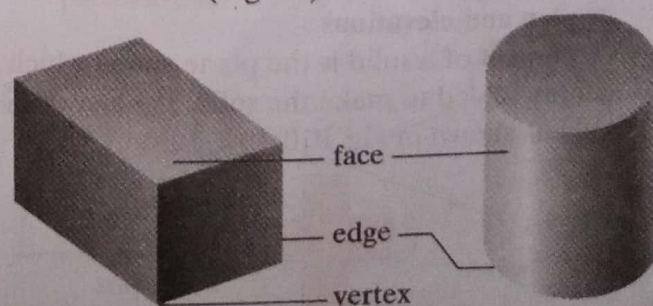


Fig. P8

Formulae for the **surface area** and **volume** of common solids are given in Table P1.

	surface area	volume
cube edge s	$6s^2$	s^3
cuboid length l , breadth b , height h	$2(lb + bh + lh)$	lbh
prism height h , base area A		Ah
sphere radius r	$4\pi r^2$	$\frac{4}{3}\pi r^3$
cylinder base radius r , height h	$2\pi r(r + h)$	$\pi r^2 h$
cone base radius r , height h , slant height l	$\pi r(r + l)$	$\frac{1}{3}\pi r^2 h$

Table P1

Fig. P9 shows two common methods of **drawing** a cuboid. These methods may be used with any solid.

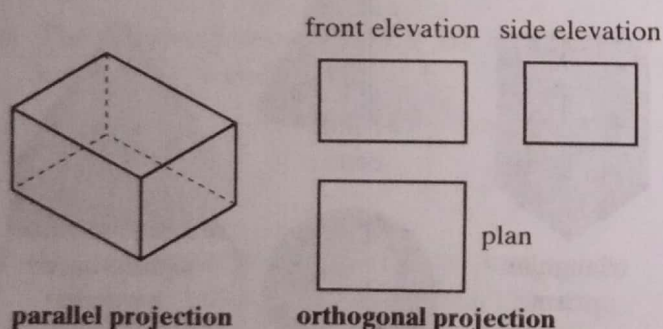


Fig. P9

In **parallel projections**, lines that are parallel on the solid appear parallel on the drawing. In **orthogonal projections**, the solid is represented by separate scale drawings of its **plan** and **elevations**.

The **net** of a solid is the plane shape which can be folded to make the solid. The net of a cube is shown in Fig. P10.



Fig. P10

b **Angle** is a measure of rotation or turning.

1 **revolution** = 360 **degrees** (1 rev = 360°)

1 **degree** = 60 **minutes** ($1^\circ = 60'$)

The names of angles change with their size. See Fig. P11.

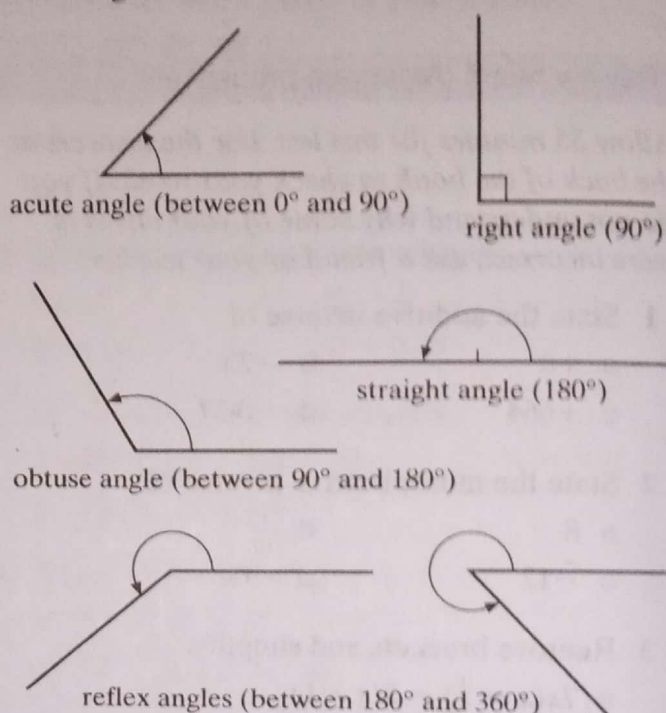
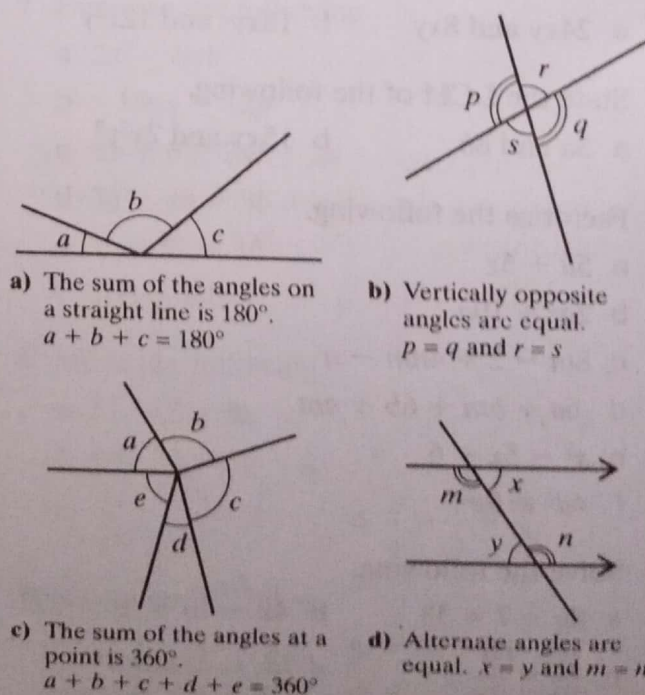


Fig. P11

Angles are measured and constructed using a **protractor**.

Fig. P12 shows some properties of angles formed when straight lines meet.



- e) Corresponding angles are equal. $a = b$ and $p = q$

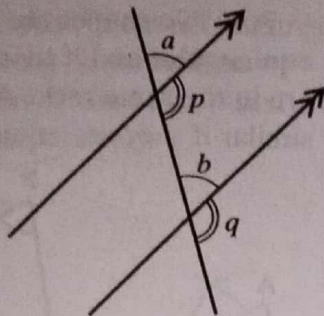


Fig. P12

In Fig. P13, α is the **angle of elevation** of the top of the flag-pole from the girl and β is the **angle of depression** of the girl from the boy.

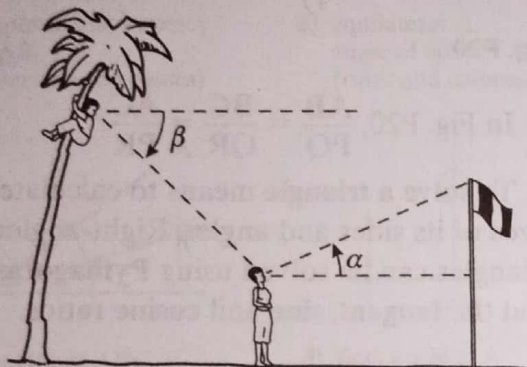


Fig. P13

Directions are taken from the **points of the compass** (Fig. P14).



Fig. P14

A **3-figure bearing** is a direction given as the number of degrees from north measured in a clockwise direction. See Fig. P15.

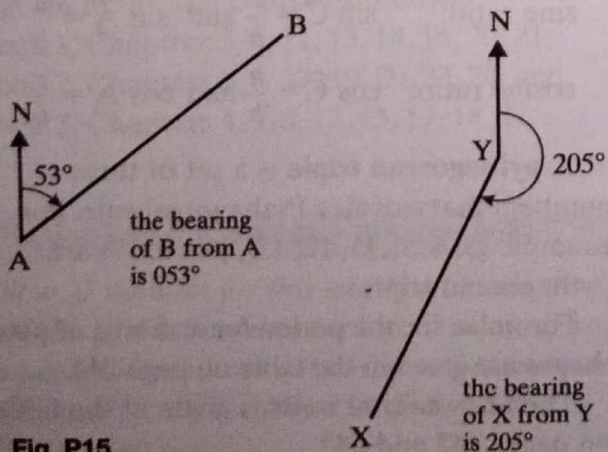


Fig. P15

The bearings in Fig. P15 can also be given as **compass bearings**: N53°E and S25°W respectively.

- c Fig. P16 shows the names and properties of some common **triangles**.

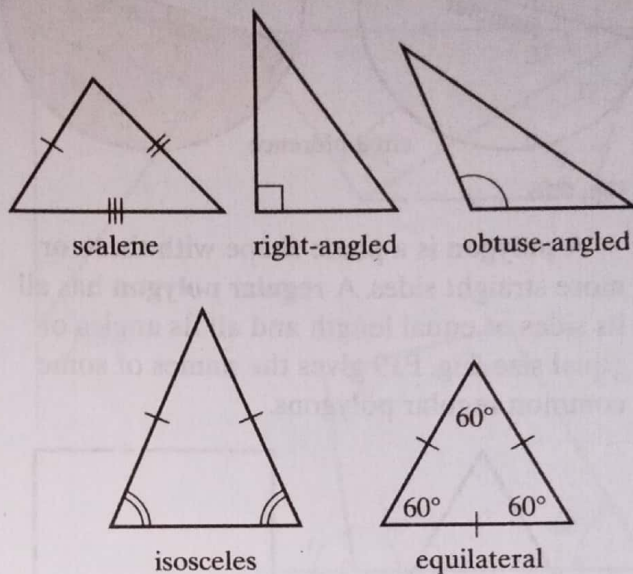


Fig. P16 Some common triangles

Fig. P17 shows the names and properties of some common **quadrilaterals**.

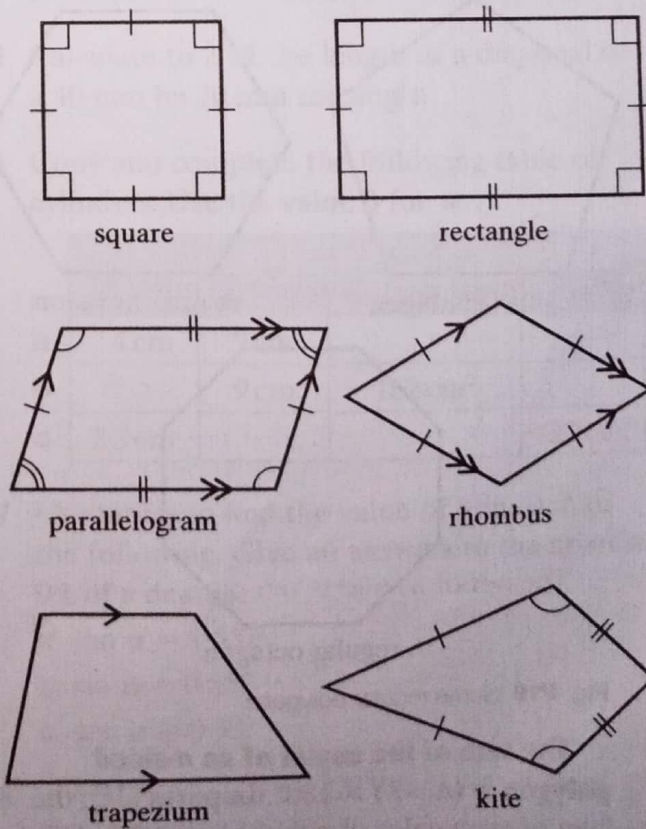


Fig. P17 Some common quadrilaterals

Fig. P18 gives the names of lines and regions in a **circle**.

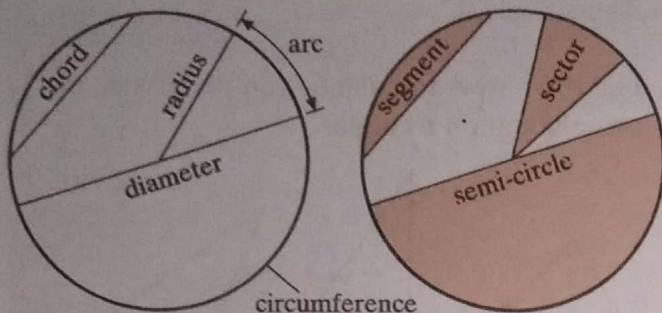


Fig. P18

A **polygon** is a plane shape with three or more straight sides. A **regular polygon** has all its sides of equal length and all its angles of equal size. Fig. P19 gives the names of some common regular polygons.

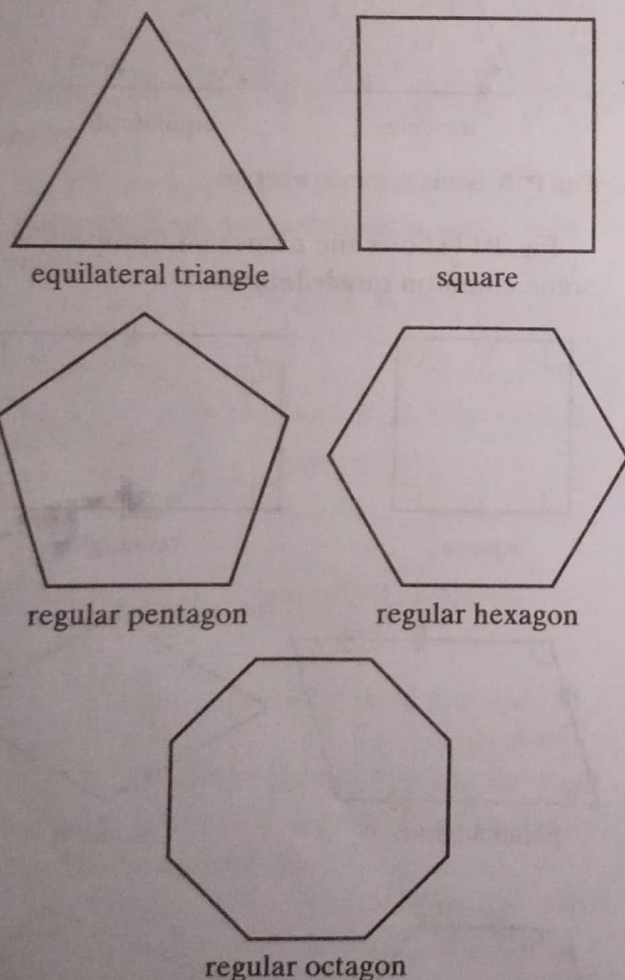


Fig. P19 Some regular polygons

The **sum of the angles of an n -sided polygon** is $(n - 2) \times 180^\circ$. In particular, the sum of the angles of a triangle is 180° and the sum of the angles of a quadrilateral is 360° .

Any two shapes are **similar** if they are **equiangular** and if their corresponding sides are in the same ratio. Any two triangles are similar if they are equiangular (Fig. P20).

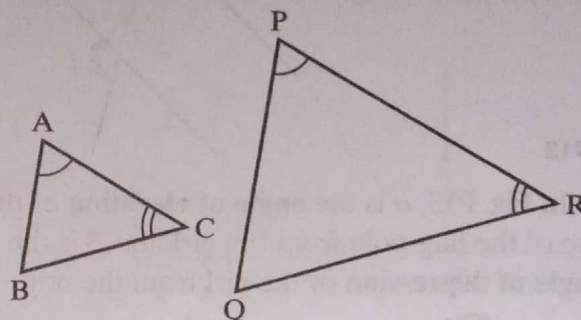


Fig. P20

$$\text{In Fig. P20, } \frac{AB}{PQ} = \frac{BC}{QR} = \frac{AC}{PR}$$

To **solve a triangle** means to calculate the sizes of its sides and angles. Right-angled triangles can be solved using **Pythagoras' rule** and the **tangent, sine and cosine ratios**.

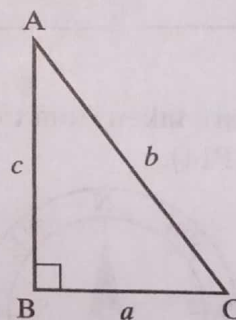


Fig. P21

In Fig. P21, $\triangle ABC$ is right-angled at B; side $\overline{AC}$ is the **hypotenuse**.

$$\text{Pythagoras' rule: } b^2 = a^2 + c^2$$

$$\text{tangent ratio: } \tan \hat{C} = \frac{c}{a} \text{ and } \tan \hat{A} = \frac{a}{c}$$

$$\text{sine ratio: } \sin \hat{C} = \frac{c}{b} \text{ and } \sin \hat{A} = \frac{a}{b}$$

$$\text{cosine ratio: } \cos \hat{C} = \frac{a}{b} \text{ and } \cos \hat{A} = \frac{c}{b}$$

A **pythagorean triple** is a set of three numbers that satisfies Pythagoras' rule. For example $\{3, 4, 5\}$, $\{5, 12, 13\}$, $\{8, 15, 17\}$ are pythagorean triples.

Formulae for the **perimeter** and **area of plane shapes** are given in the table on page 244.

The **SI system of units** is given in the tables on pages 242 and 243.

- d The sketches in Fig. P22 show the main features of the common **geometrical constructions**.

To construct angles of 45° and 30° , bisect angles of 90° and 60° respectively.

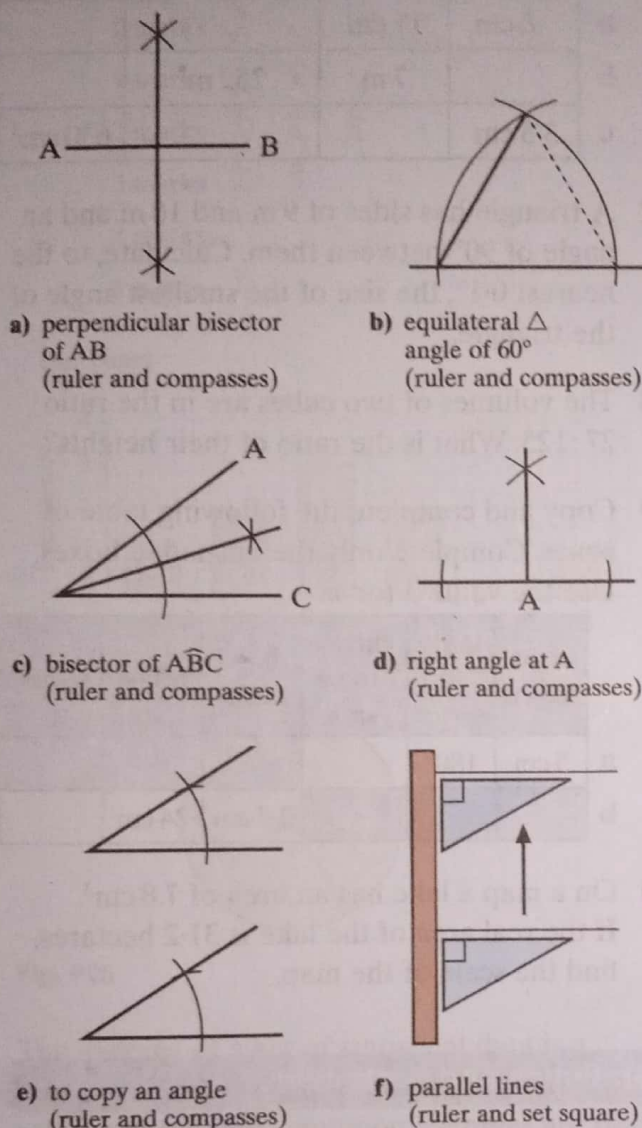


Fig. P22

Detailed coverage of geometry and mensuration is given in the JSS course as follows:

Book 1, Chapters 3, 6, 11, 13, 14, 16; 20, 21;
Book 2, Chapters 3, 6, 17, 19, 20, 23, 24; and
Book 3, Chapters 4, 7, 8, 12, 15, 17, 18, 22.

Review test 5 (Geometry and mensuration)

Allow 35 minutes for this test. Use the answers at the back of the book to check your work. If you do not understand why some of your answers were incorrect, ask a friend or your teacher. Then try Test 6.

- One of the angles of a parallelogram is 37° . What are the sizes of its other three angles?
- How many lines of bilateral symmetry does a square have?
- Find the values of m, n, p, q, r, s in Fig. P23.

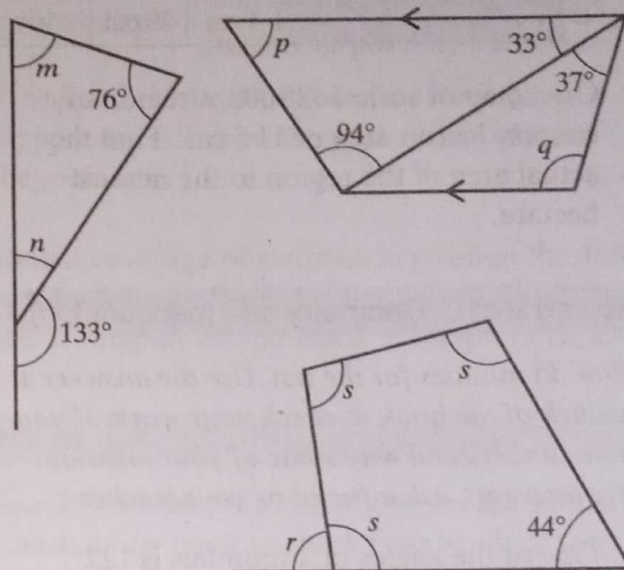


Fig. P23

- Which one of the following is a Pythagorean triple?
a 4, 5, 6 b 5, 13, 14 c 6, 8, 10
- Calculate to 2 s.f. the length of a diagonal of a 40 mm by 70 mm rectangle.
- Copy and complete the following table of cylinders. Use the value 3 for π .

	radius	height	curved surface area	volume
a	4 cm	9 cm		
b		9 cm	162 cm^2	
c	2.5 cm			120 cm^3

- Use tables to find the value of α in each of the following. Give all answers to the nearest 0.1 of a degree.
a $\tan \alpha = 1.552$
b $\sin \alpha = 0.555$
c $\cos \alpha = 0.33$
- $\triangle ABC$ is similar to $\triangle XYZ$. If $AC = 9 \text{ cm}$, $BC = 5 \text{ cm}$ and $YZ = 7 \text{ cm}$, find the length of XZ .

- 9 Copy the following table of cones. Complete only the unshaded boxes. Use the value 3 for π .

	slant height	angle of sector	curved surface area	base radius	height	volume
a	8 cm	210°				
b				4 cm	20 cm	

- 10 On a map of scale 1 : 25 000, a region of a country has an area of 11.5 cm². Find the actual area of the region to the nearest hectare.

Review test 6 (Geometry and mensuration)

Allow 35 minutes for the test. Use the answers at the back of the book to check your work. If you do not understand why some of your answers were incorrect, ask a friend or your teacher.

- One of the angles of a rhombus is 122°. What are the sizes of its other three angles?
- How many lines of bilateral symmetry does an equilateral triangle have?
- Find the values of p, q, r, s, t, x in Fig. P24.

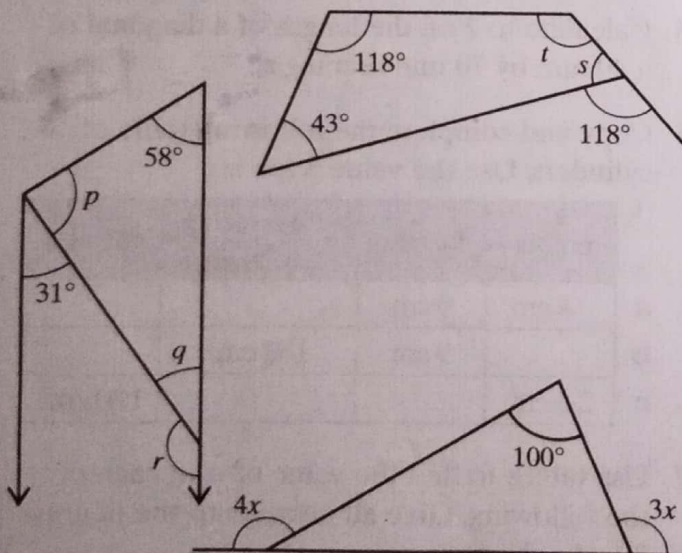


Fig. P24

- Two of the sides of a right-angled triangle are 3 m and 4 m. Find the size of the third side. [There are two possible answers.]
- A chord is 3 cm from the centre of a circle of radius 5 cm. What is the length of the chord?

- 6 Copy and complete the following table of cylinders. Use the value 3 for π .

	radius	height	curved surface area	volume
a	6 cm	11 cm		
b		7 m	252 m ²	
c	3.5 cm			630 cm ³

- A triangle has sides of 9 m and 10 m and an angle of 90° between them. Calculate, to the nearest 0.1°, the size of the smallest angle of the triangle.
- The volumes of two cubes are in the ratio 27 : 125. What is the ratio of their heights?
- Copy and complete the following table of cones. Complete only the unshaded boxes. Use the value 3 for π .

	slant height	angle of sector	curved surface area	base radius	height	volume
a	5 cm	180°				
b				2.5 cm	24 cm	

- 10 On a map a lake has an area of 7.8 cm². If the real area of the lake is 31.2 hectares, find the scale of the map.

Statistics and probability

- a Information in numerical form is called **statistics**. Statistical **data** may be given in **rank order** (i.e. in order of increasing size) such as in the following marks obtained in a test out of 5:

0, 1, 1, 2, 2, 2, 3, 3, 5

Data may also be given in a **frequency table** (Table P2).

mark	0	1	2	3	4	5
frequency	1	2	3	2	0	1

Table P2

The **frequency** is the number of times each piece of data occurs.